Partha Chakraborty

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SUMMARY

Software engineer with 2+ years at startups building API integrations, workflow automation, and LLM features on Python, FastAPI, Node.js, and React, plus a deployed multi-agent RAG app. M.S. in AI and Machine Learning, University at Buffalo, expected January 2027.

EXPERIENCE

TurboMail **Remote**
AI Engineer Intern *May 2026 – Aug 2026*

- Wrote bulk email deletion and folder-move APIs in FastAPI, plus email-body sanitization and multi-account thread scoping.
- Automated unsubscribe handling with FastAPI background tasks that parse RFC 8058 List-Unsubscribe headers.
- Designed the database schema and APIs for subscription and payment tracking, then built the customer-facing React interface.
- Owned production features across the stack in React, TypeScript, Python, FastAPI, SQLAlchemy, and SQL.
- Fine-tuned Qwen2.5-7B with LoRA and PEFT in PyTorch on a custom email dataset for an in-product LLM feature.

Bird Vision AB **Remote**
Software Developer *Jun 2023 – Jul 2025*

- Integrated Eboka with the company's other products end to end, so customers got value from each in one subscription cycle.
- Built two-way customer data sync between two company apps, with scheduled background bulk imports that notify users.
- Added on-demand single-customer import by user ID in both directions, so users didn't have to enter the same data twice.
- Took backend features to production, connecting third-party APIs to existing workflows with Node.js and Express.js.
- Ranked second on the team in code contributions within a year, shipping frontend and backend work across three products.
- Wrote Cypress tests covering the main user flows in the web app.

EDUCATION

University at Buffalo **New York, USA**
M.S. in Artificial Intelligence and Machine Learning *GPA 3.9/4.0 | Expected Jan 2027*
Coursework: Algorithms, Database Systems, Data-Intensive Computing, Machine Learning

Presidency University **Bengaluru, India**
B.Tech. in Computer Science and Engineering *GPA 8.14/10 | 2018 – 2022*
Coursework: Computer Networks

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, HTML, CSS
Backend and integrations: FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, background tasks, third-party APIs
AI and LLMs: Gemini API, multi-agent RAG, tool calling, embeddings, vector search, BM25, reranking, PyTorch, LoRA, PEFT
Data and tools: PostgreSQL, MySQL, MongoDB, Firebase, ChromaDB, Docker, Linux, Git, Pandas, Streamlit
Frontend and testing: React, Next.js, Redux, Tailwind CSS, React Native, Cypress
AI coding tools: Cursor, Claude Code, GitHub Copilot

PROJECTS

AI Codebase Intelligence [Live Demo] [GitHub] Python, Gemini, ChromaDB, Streamlit

- Deployed a multi-agent RAG app on the Gemini API that answers questions about public GitHub and local repositories through four agents: Architect, Code Reviewer, Bug Finder, and Documentation.
- Grounded answers in source code with Tree-sitter chunking across six languages, hybrid semantic, BM25, and symbol retrieval, and cross-encoder reranking.
- To keep answers reliable, built a source explorer with file paths, line ranges, and citation checks; reached 87.5% Hit@5 and 0.750 MRR on an initial 8-question retrieval evaluation.

GraphBench [Live Demo] Python, NetworkX, LadybugDB, Streamlit, Altair

- Built a software dependency analysis tool that uses PageRank, betweenness centrality, cycle detection, and K-core analysis to find critical dependencies, bottlenecks, and cycles.
- Guided CSV ingestion confirms edge direction and validates data before loading, and an interactive explorer with JSON and CSV reports explains findings in plain language.
- Simulates failures by tracing direct and indirect dependents, blast radius, impact depth, and shortest dependency paths.

CERTIFICATIONS

Hugging Face Agents Course, Unit 1: Fundamentals of Agents Oct 2026